

A \$12 million battery. A \$57 million project.

Hyde Park BESS (Project 21334): where the cost of a utility-owned, distribution-connected battery storage station actually sits, and how it compares to industry benchmarks

Cost Estimating Center of Excellence | Estimate basis: ESF Total Rev 3, 24 Sep 2026 | 29 Sep 2026

Internal. Contains vendor-confidential pricing.

THE PROMISE

By the end of this deck you can answer one question in one breath

“Why is a \$12 million battery a \$57 million project?” Because the battery is the only part a vendor prices. The station, the site, the feeder protection, the siting case, the risk and the cost of money are ours.

\$12.1M

Vendor fixed-price contract

Nomad cover agreement, 10 Mar 2026.
12 Voyager LFP units, PCS, BMS, fire protection, 10-yr warranty. \$605/kWh, \$1,211/kW.

\$57.2M

Project 21334 approval level

ESF Total Rev 3. Station 360 \$50.0M plus Station 496 protection \$7.3M. \$2,862/kWh, \$5,725/kW. Conceptual, -25%/+50%.

21%

Battery share of the total

Add the D-Line circuit extension (24211, \$4.8M conceptual) and the share drops to 20%. Four dollars of every five are not the battery.

What this deck contributes: a line-by-line extraction of the Rev 3 estimate into thirteen cost layers, unit costs at every boundary a benchmark could be quoted at, the cost history from the 2021 IFR to today, and the reconciliations that must close before full funding.

What a vendor quote contains, and what it leaves for the utility

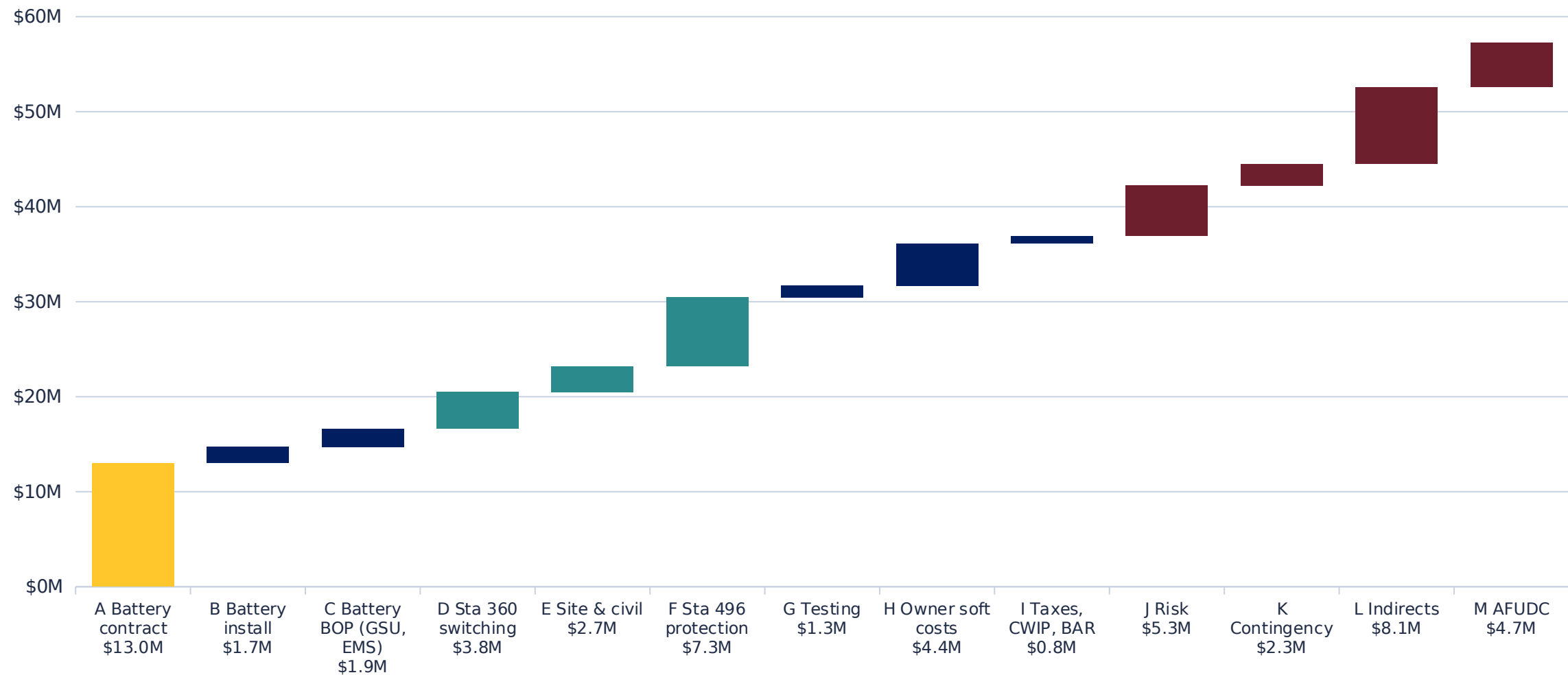
Inside the \$12.1M Nomad price

- 12 containerized LFP units, 10 MW / 20 MWh, 12 MVA
- Inverters (PCS), BMS, HVAC, fire detection and suppression inside the container
- Spares \$125K, shipping \$90K, remote support \$210K
- 10-year warranty; years 3-7 extension \$798K
- Field assembly supervision, commissioning, training: \$246K
- Fixed until Final Acceptance; LDs capped at 10%

Outside it: \$45.1M, 79% of the project

- **Station 360**, \$3.8M direct: four 15 kV breaker skids, GSUs, SSVT, metering, cable, duct bank, ground grid
- **Site**, \$2.7M direct: a 200 ft x 200 ft platform on a closed landfill, access road, retaining wall
- **Station 496**, \$7.3M: 20 relays, transfer trip, RTU work so four feeders can accept 2.5 MW each
- **Owner scope**, \$10.1M: install labor, GSUs, EMS, testing, engineering, Article 80 siting, outreach, PM, taxes
- **Station 360 risk, contingency, indirects, AFUDC**, \$20.3M: 35% of the total (Station 496 carries its own inside its \$7.3M)

Thirteen layers between the battery contract and the approval level



Gold = battery contract as carried in Rev 3 (\$12.1M contract plus \$1.1M escalation on the unpaid balance; see slide 12). Teal = station, site and feeder protection. Burgundy = risk, contingency, indirects, AFUDC. Total \$57.2M.
EstimatingCoE | Eversource Energy | Stewards of Affordability

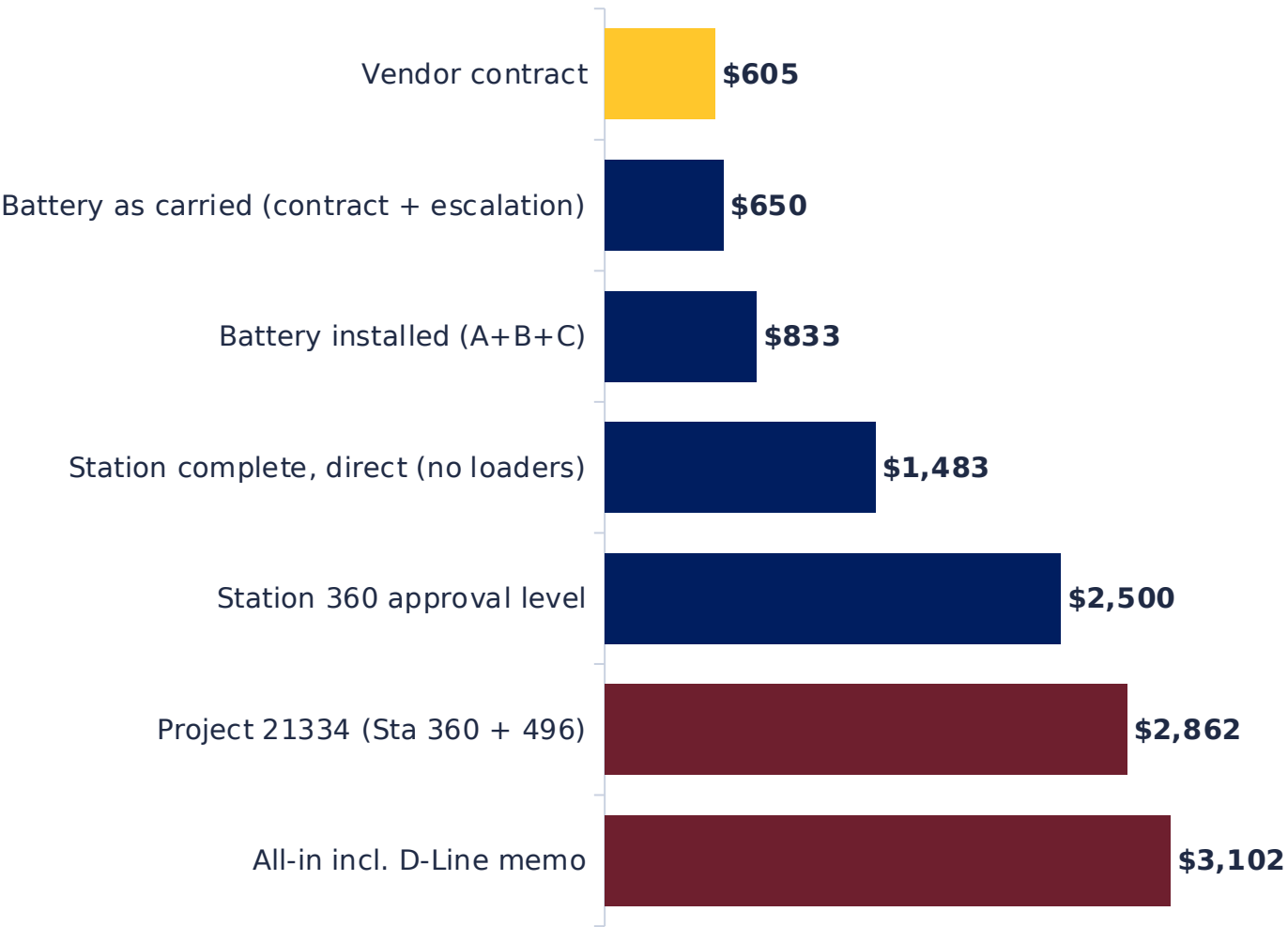
LOADED VIEW: EVERY LAYER CARRIES ITS SHARE OF THE LOADERS (FACTOR 1.685 ON STATION 360 DIRECTS)

Where the \$57.2M sits when risk, contingency, indirects and AFUDC are spread

Layer	Direct	Loaded	Share
Battery supply contract (Nomad)	\$13.0M	\$21.9M	38%
Battery installation & integration (owner scope)	\$1.7M	\$2.9M	5%
Battery balance-of-plant (GSUs, EMS/comms, aux power)	\$1.9M	\$3.3M	6%
Station 360 13.8 kV switching & interconnection facilities	\$3.8M	\$6.5M	11%
Site development & civil (former landfill site)	\$2.7M	\$4.5M	8%
Station 496 feeder protection upgrades (interconnection)	\$7.3M	\$7.3M	13%
Testing & commissioning	\$1.3M	\$2.1M	4%
Owner soft costs (engineering, siting, environmental, outreach, PM, land)	\$4.4M	\$7.4M	13%
Taxes, CWIP, builders-risk insurance	\$0.8M	\$1.3M	2%
Total, project 21334 (directs incl. \$20.3M of loaders J-M)	\$57.2M	\$57.2M	100%

Battery-related layers A+B+C loaded: \$28.1M (49%). Station, site and feeder interconnection D+E+F loaded: \$18.3M (32%). Owner soft costs, testing and taxes: \$10.9M.

The same project is \$605/kWh or \$2,862/kWh depending on where you draw the line

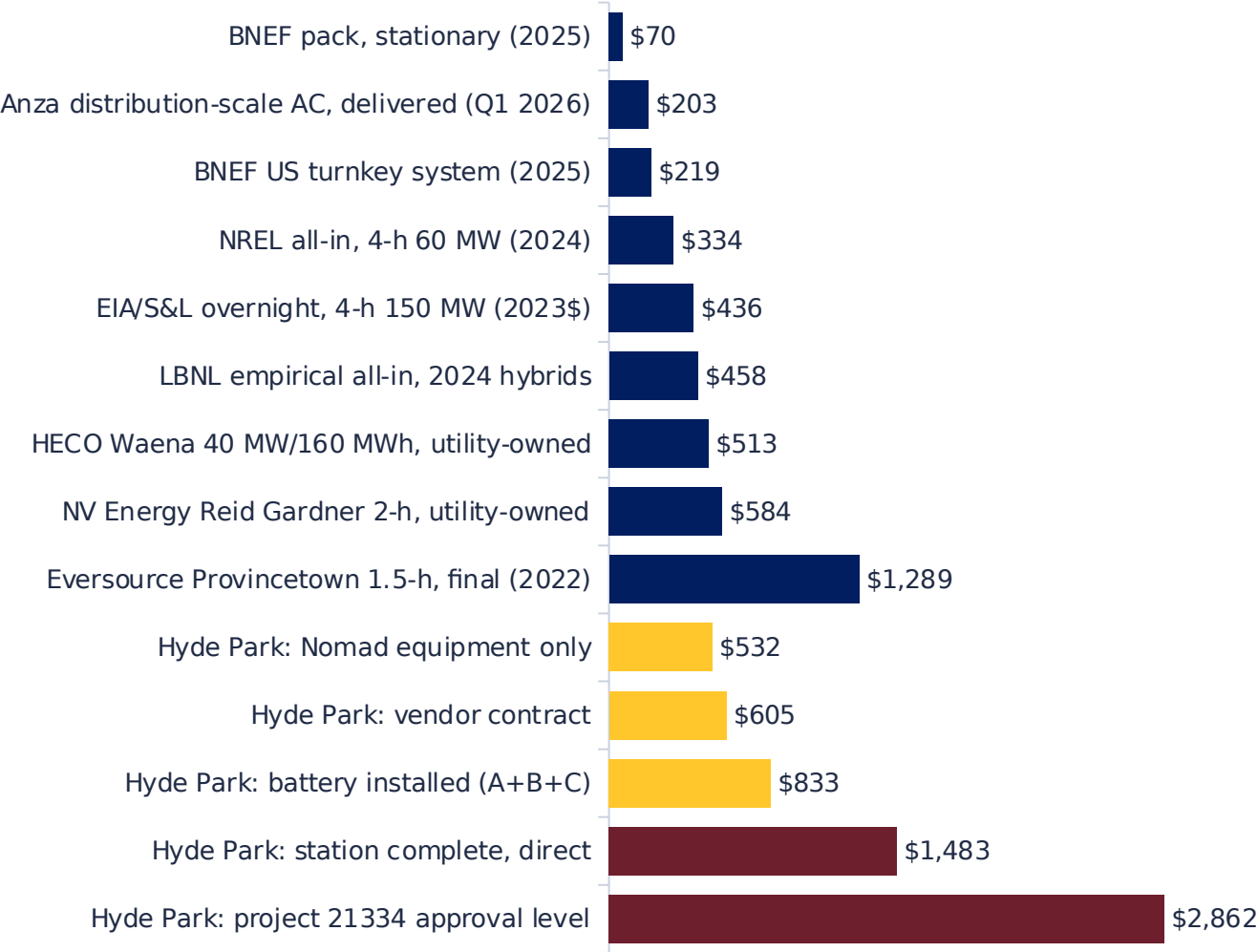


Boundary	\$/kW	\$/kWh	x
Vendor contract	\$1,211	\$605	1.0x
Battery installed	\$1,666	\$833	1.4x
Sta 360 approval	\$4,999	\$2,500	4.1x
Project 21334	\$5,725	\$2,862	4.7x

Incl. D-Line \$6,204 \$3,102 5.1x

A benchmark is only meaningful against the row that matches its definition. Pack and DC-block prices belong beside the top bar; NREL and EIA installed costs beside 'battery installed' or 'station complete'; nothing published includes a utility's own indirects and AFUDC.

The battery is at market; the project is above it, for reasons the benchmarks exclude



- **Battery at market.** Nomad's \$532/kWh equipment price sits between BNEF's US turnkey (\$219, 4-h, 100 MW class) and the disclosed utility-owned projects (\$513-584). For a 12-unit, 2-h, FEOC-era order in New England that is a fair price.
- **Installed battery (A+B+C) at \$833/kWh** is above every national all-in index because those indices are four-hour and 60-150 MW. NREL's own cost structure (pack \$/kWh x hours + BOS \$/kW) puts a 2-h system at roughly 1.3-1.5x the 4-h \$/kWh.
- **The project at \$2,862/kWh** is not a battery benchmark at all. It carries a switching station, feeder protection at Station 496, a landfill platform, Article 80 siting, indirects and AFUDC. The closest published analogue is Provincetown at \$1,289/kWh (2022 dollars, no separate station, existing site).
- **No published index includes a utility's indirects or AFUDC.** Strip J-M and Hyde Park is \$1,483/kWh direct; strip the station and site layers D and E as well and it is \$1,156/kWh.

Six things about Hyde Park that a 100 MW greenfield benchmark never has to pay for

Size. 10 MW / 20 MWh. A switching station, a siting case and a commissioning team cost about the same at 10 MW as at 100 MW; per kWh they are ten times heavier.

Duration. Two hours. Fixed \$/kW items (PCS, breakers, GSUs, interconnection) are spread over half the MWh of a four-hour benchmark.

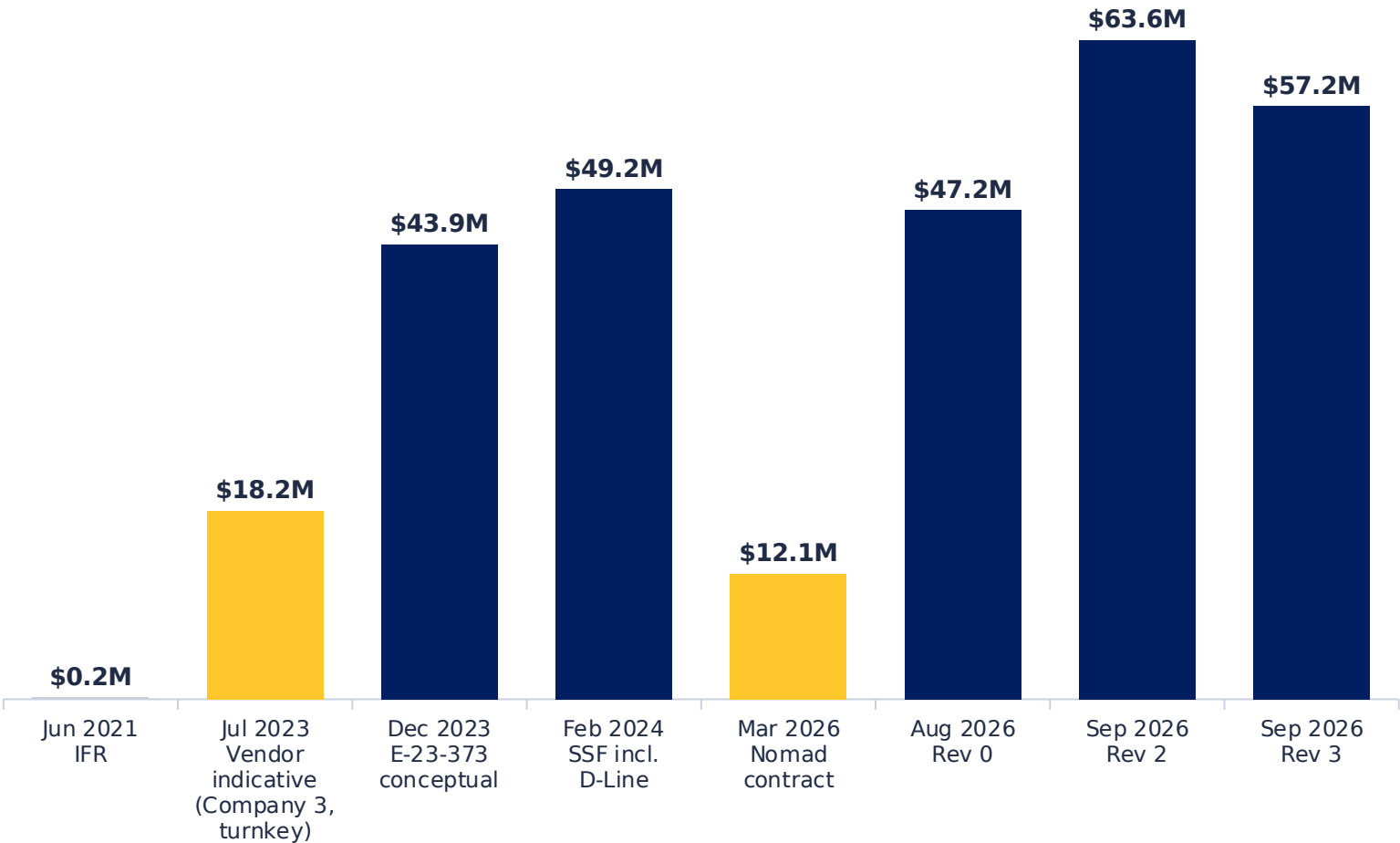
Site. A closed construction-debris landfill in Boston: DEP post-closure use permit, cap protection, 36 borings, imported platform fill, 14 ft retaining wall, unknown obstructions (\$450K risk).

Interconnection. Four feeder tie-ins at 13.8 kV, each with its own breaker skid, plus relay replacement and transfer trip at Station 496 (\$7.3M) and a 5,465 ft circuit extension (24211).

Siting and outreach. Article 80 Small Project Review, Zoning Board of Appeal, outside counsel, an EJ community, a neighborhood concerned about open land and views: \$1.5M plus \$0.5M in the risk register.

Utility accounting. Corporate indirects (\$10.0M) and AFUDC over a seven-year development window (\$5.0M) are real costs to customers that a developer's turnkey price never shows.

Five years, five in-service dates, and an estimate that moved from \$44M to \$57M

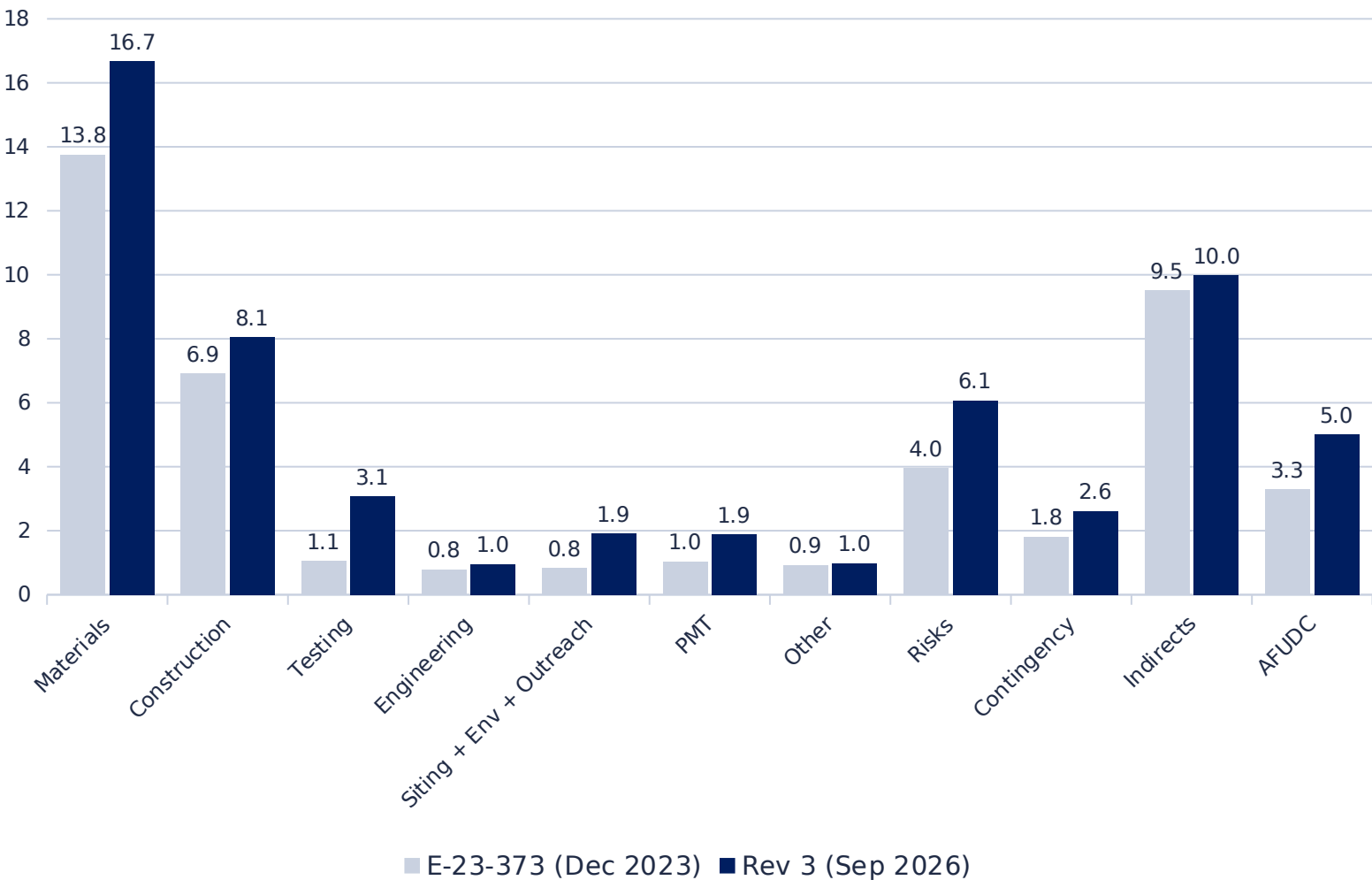


In-service date drift

As of	Planned ISD
Jun 2021	Q3 2025
Dec 2023	Apr 2027
Feb 2024	Sep 2027
Jan 2026	Aug 2028
Jun 2026	Dec 2028

Gold bars are vendor prices for the battery scope only. Navy bars are full project estimates. The battery price fell from \$18.2M turnkey-indicative (2023, incl. civil) to \$12.1M contracted (2026); the project rose \$13.3M in the same window. The 2023 conceptual carried a placeholder that was never re-based until August 2026.

What changed between the December 2023 conceptual and Rev 3: \$43.9M to \$57.2M

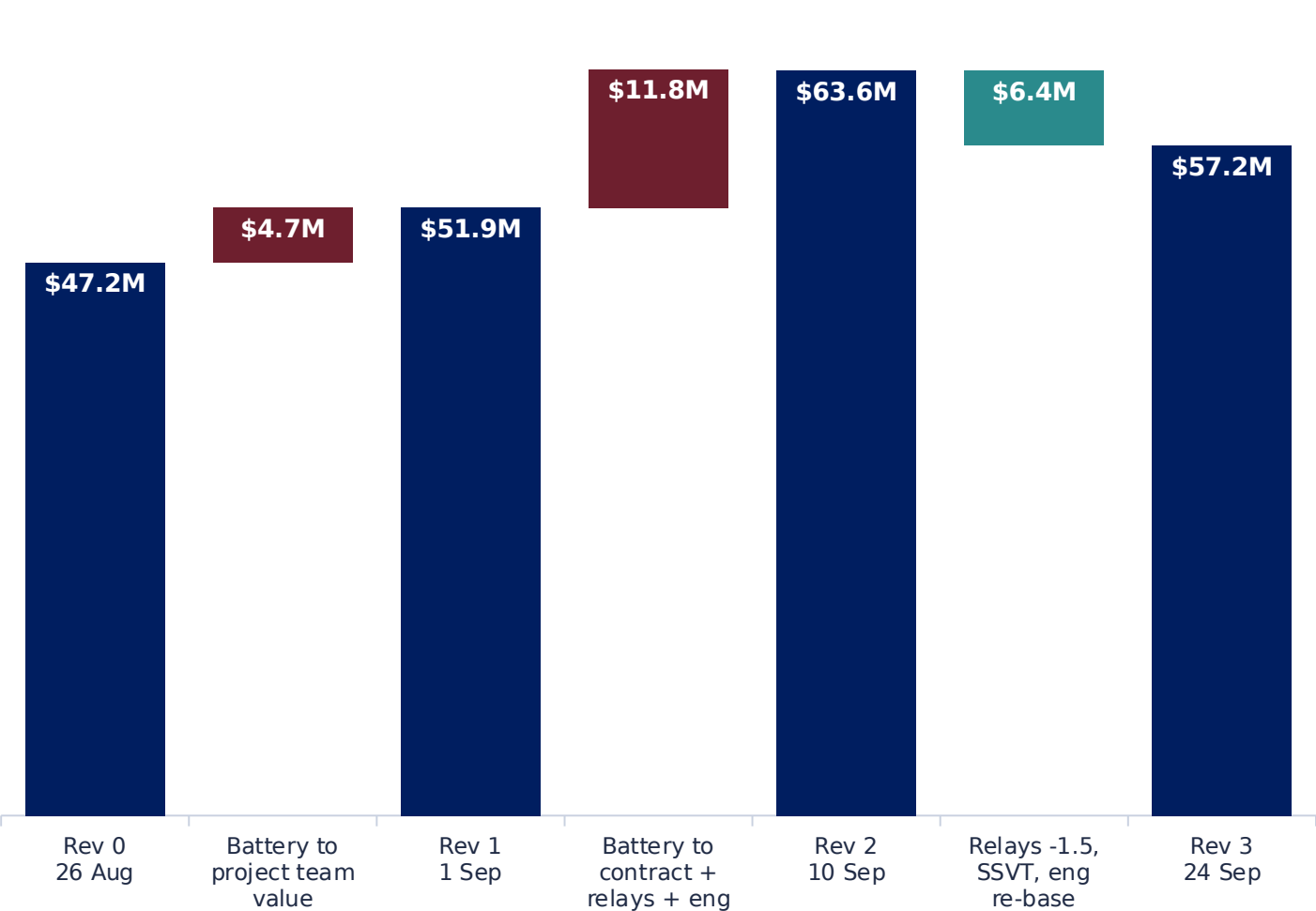


Largest movements

Category	Delta
Materials	+\$2.9M
Risks	+\$2.1M
Testing	+\$2.0M
AFUDC	+\$1.7M
Construction	+\$1.1M
Siting + Env + Outreach	+\$1.1M
PMT	+\$0.9M

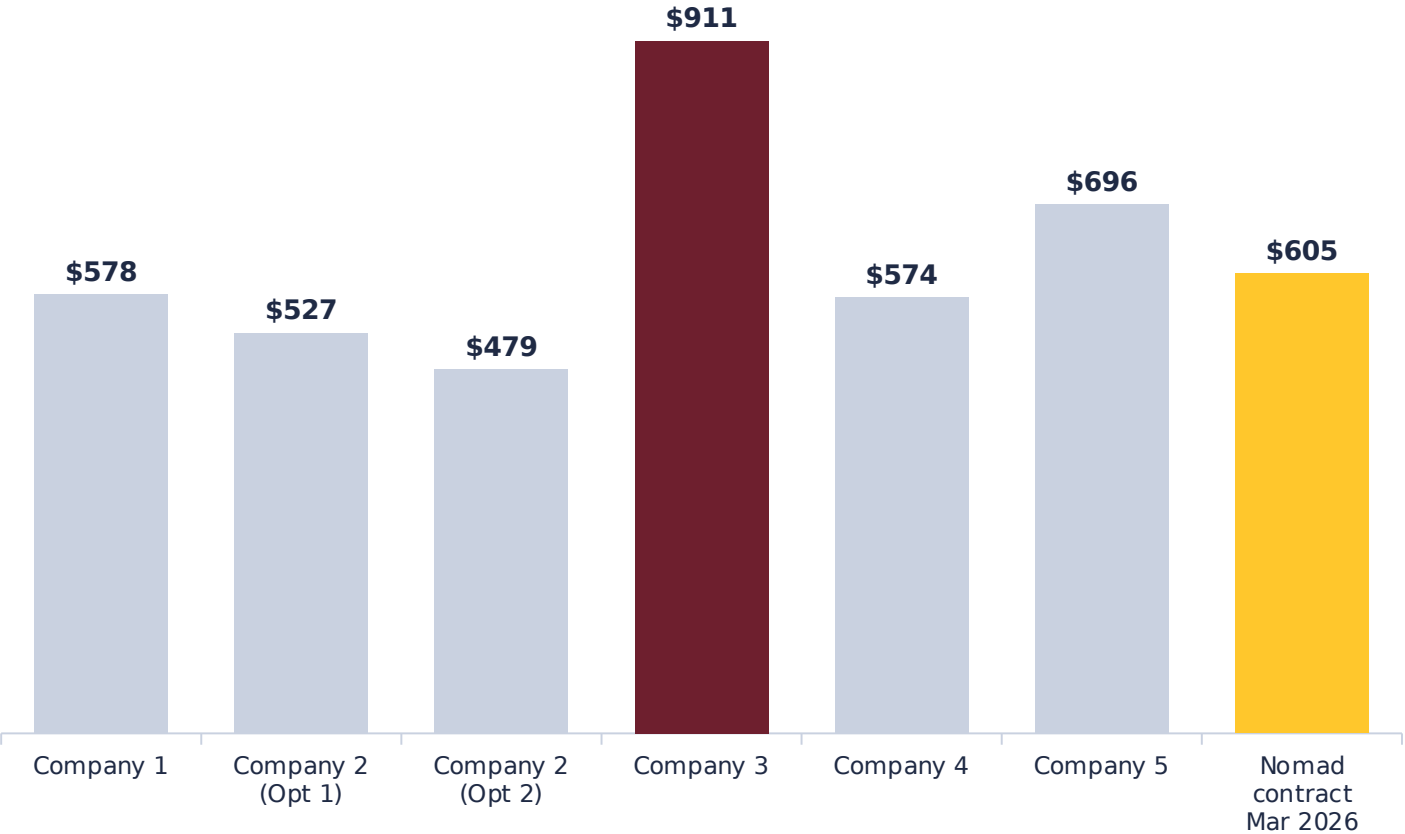
Materials: contracted battery replaces 2023 indicative pricing, plus breaker skids and GSUs. Testing and siting: Station 496 scope and Article 80 were not in E-23-373. AFUDC: a 20-month later ISD on a larger base.

Four revisions in five weeks: how the September estimate settled at \$57.2M



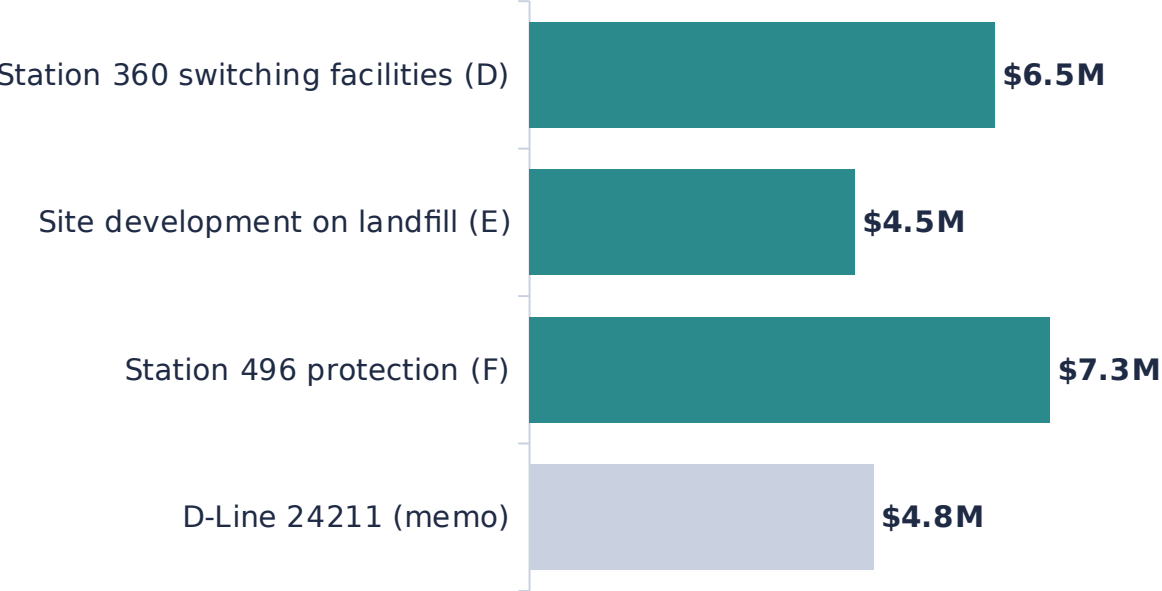
- **Rev 0 carried the battery at about \$6M** from the 2023 indicative sheet. The PM: 'we know the BESS system itself is 12.'
- **Rev 2 reset it to the cover agreement, escalated to \$13.1M**, and carried \$1.5M of 'relays for batteries' from a 2023 bidder that priced its own switchgear.
- **Rev 3 removed the relays** (P&C Engineering: included in the \$300K Siemens skid), cut three SSVTs to one, and re-based engineering to \$501K to go plus actuals.
- **Net:** +\$10.0M from Rev 0, of which \$7.0M is the battery line catching up to the contract.

The battery itself: a \$9.6M-\$18.2M indicative spread in 2023, a \$12.1M fixed price in 2026



- **July 2023, five bidders, 10 MW / 20 MWh:** \$9.6M to \$18.2M. Every bidder excluded interconnection, permits beyond local, below-grade risk, sound walls or spares.
- **Company 3 (\$18.2M, burgundy)** became the E-23-373 basis because it alone priced turnkey civil (\$4.56M) and installation.
- **March 2026 contract, \$12.1M (gold):** supply only, with supervision. Civil, install, GSUs, station and interconnection moved to Eversource scope.
- **Equipment-only inside the contract:** \$10.64M, \$532/kWh, in line with published 2025-26 turnkey system indices (slide 7).
- **Rev 3 adds \$1.13M of escalation** to a price the contract fixes until Final Acceptance.

Station and interconnection facilities: \$18.3M loaded, and none of it in a vendor quote



- **Four Siemens SDV7 breaker skids at \$300K each** (relays, PTs, DC battery, cabinet on the skid), disconnects, four metering transclosures, station service, riser poles, 15 kV cable and 1,240 ft of duct bank.
- **Station 496:** 20 SEL relays replacing SEL-551 and REF610 units on four feeders, RTU and RTAC points, transfer trip over dark fiber; \$1.8M of that is testing.
- **Site:** 2.5 acres cleared, 760 ft road, 2,520 CY cut, imported fill, 14 ft retaining wall, ground grid, fence, lighting; Haugland's platform take-off is larger than ours and is unreconciled.
- **D-Line (24211):** 5,465 ft feeder extension, three manholes, 16,395 ft of 3-700 cable; \$4.8M is a 2023 conceptual figure.

\$18.3M

Loaded, in 21334

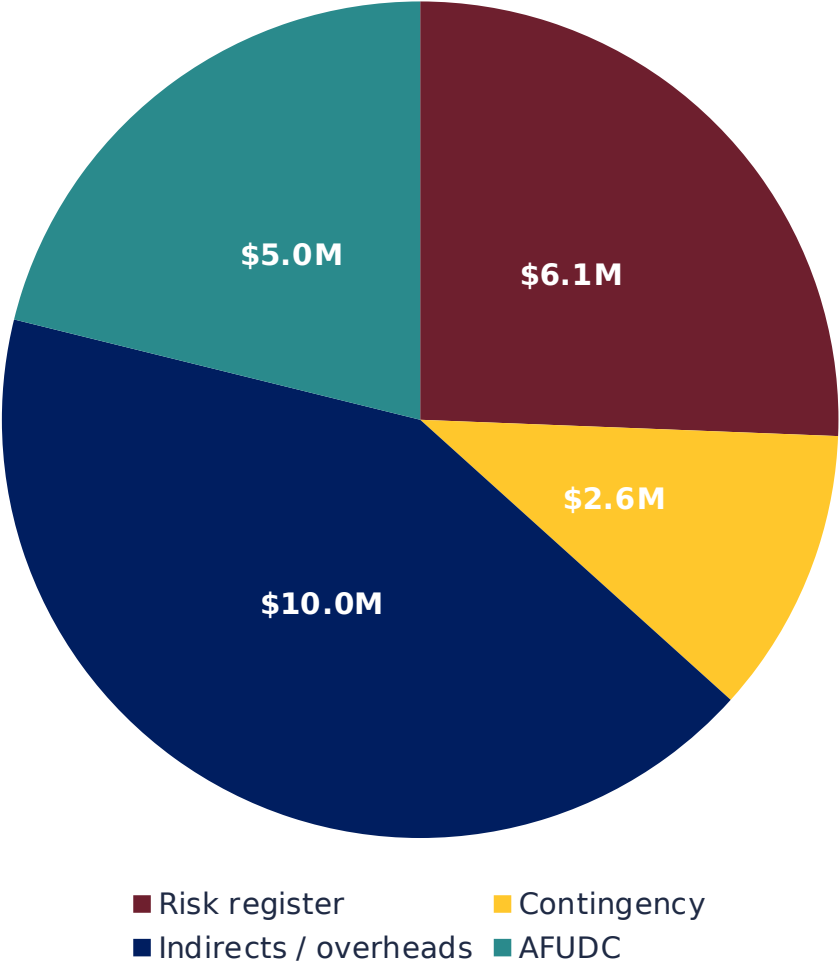
32% of the approval level

\$23.1M

Incl. D-Line memo

37% of the all-in view

Risk, contingency, indirects and AFUDC are 41% of the approval level



Station 360 risk item	\$K
Material cost, tariffs	1,000
Design / scope change orders	800
Labor cost, shortage, overtime	800
Unforeseen UG obstructions (landfill)	450
Environmental (DEP, landfill closure docs)	300
Soil contamination remediation	300
Station and P&C engineering changes	600
Weather, siting delay, outreach, BAR deductibles	1,050

Risk plus contingency is 17.9% of the approval level before loaders, inside the 15-18% lead-review guidance. Indirects at 17.4% and AFUDC at 8.8% follow the EMA station template on a spend curve that runs from July 2021 to December 2028. Haugland's Attachment 4-F carries overlapping exposures (geotech \$0.6M-\$3.5M, fault-study equipment change \$0.25M-\$1.8M) that have not been reconciled with this register.

What must close before this number goes to EPAC as full funding

Item	Type	Dollars at stake	Owner
Reconcile Haugland Class 3 base (\$19.2M, its scope) to Rev 3 layers B+D+E (\$8.3M direct). Red team expects it to move the base up, not down.	Scope boundary	+\$4M to +\$8M direct if half the gap is 21334 scope	Cost Estimating + PM + HEG
Siting chain has no float: ZBA determination Feb 2028, construction start Feb 2028, ISD Dec 2028; EFSB fallback 18-24 mo	Schedule	~\$5M per 12 months (AFUDC, escalation, PM, indirects)	Siting + PM
Present full funding at its real maturity: 30% engineering, no bid, placeholder platform = Class 4	Classification	Range is -25/+50% (\$43M-\$86M) until closed	Cost Estimating
Geotechnical investigation and NFPA 855 hazard mitigation analysis before civil freeze	Study	\$1.0M of deducts vs \$0.6M-\$3.5M exposure	Civil + Capital Projects Eng.
Confirm four written design-basis items (retaining wall, basin, duct bank, GSU count)	Design confirmation	\$1.47M HEG deduct ceiling (~\$0.8M net of GSUs already at 4)	Substation Engineering
Escalation on the fixed-price battery balance; check Exhibit C for tariff / change-in-law pass-through	Decision	\$1.13M	PM / Cost Estimating / Sourcing
Battery installation labor: replace 25%-of-16,801 mh with a crew build-up	Estimate	+/- \$0.5M (Rev 2 used 50%)	Cost Estimating + Nomad + HEG
Station 496 at \$363K per relay: compare to two recent EMA relay-replacement actuals; outage-window premium not priced	Benchmark	Unknown; \$1.8M testing line	P&C Engineering
Spot generation if the BESS misses Dec 2028 (SSF: 2-8 MW) was removed from scope; who carries it?	Scope / ownership	Not in project	System Planning
Re-base D-Line 24211 (2023 conceptual \$4.8M; HEG take-off larger)	Estimate	Unknown	Distribution Engineering

Defensible reductions (~\$3M: battery escalation, design-basis confirmations, material risk sized on non-battery material) are conditional on the Haugland reconciliation closing first.

Red team: three assumptions that, if wrong, move the number more than the reserve

1. Construction may be low, not high.

Rev 3 layers B+D+E are \$8.3M direct. Haugland's Class 3 base for overlapping scope is \$19.2M with a \$2.9M deduct ceiling. Nobody has mapped one to the other.

Settles it: a signed HEG-to-Rev 3 scope map before the PAF is drafted.

2. Siting is binary and carries no float.

ZBA determination, construction start and container delivery all land in Q1 2028 for a Dec 2028 ISD. Denial means an EFSB exemption at 18-24 months. The estimate carries \$250K for siting delay.

Settles it: Article 80 filing date, ZBA calendar, P6 float on the siting chain.

3. The number is Class 4 wearing Class 3 clothes.

30% engineering, unit-library construction with no bid, install labor as a percentage of a 2023 total, a 2.5 ft platform placeholder. EPAC will hear \$57.2M as a planning number.

Settles it: geotech, HMA, HEG map, crew build-up; then reclassify with a -15/+25% range.

Stress tests on the package's own numbers, and what history says about small utility batteries

Scenario	Loaded cost	Date	Covered by \$7.6M reserve?
Construction unit costs +25% on B+D+E+G	+\$4.0M	0	Yes, alone
HEG reconciliation at midpoint of the gap (+ \$5M direct)	+\$8.4M	0	No
Schedule +12 months (siting or long-lead)	~+\$5M	+12 mo	Partly
EFSB zoning exemption path	~+\$8M to + \$10M	+18-24 mo	No
Contractor bids +20% over estimate labor	+\$2.1M	0	Yes, alone
Combination: HEG midpoint + 12-month slip	~+\$13M (to ~\$70M)	+12 mo	No
Battery change orders at the LD cap (10%)	+\$1.2M	0	Yes

Untestable from the package: tariff or change-in-law pass-through on the Nomad price (Exhibit C terms), the AFUDC rate (assumed ~7% on CWIP), Station 496 outage premium (1.35 to 1.70 factor on affected hours), FEOC / ITC effect on net cost.

Base rate: approved to actual	
Project	Uplift
BGE Fairhaven 2.5 MW/9.7 MWh	+64%
HECO Waena 40 MW/160 MWh	+37%
SaskPower Regina 20 MW	+31%
Eversource Provincetown	+9% to +40%
Hyde Park, Dec 2023 to Sep 2026	+30%
Hyde Park, Rev 0 to Rev 3 (5 wks)	+21%

Rev 3 carries 17.9% risk plus contingency. The exception case for beating the base rate rests on three anchors: a fixed battery price, a Siemens price on the skids, and Haugland engaged pre-construction. Together they cover about a third of the cost. The other two thirds have no such anchor.

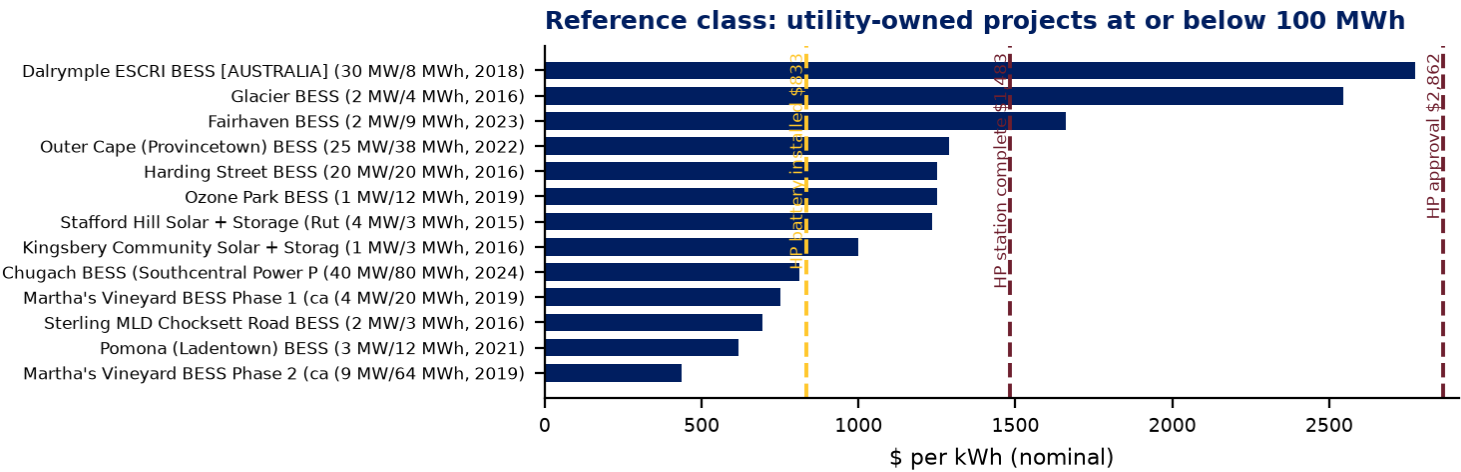
Pre-mortem: it is June 2031, the project finished at \$74M and fifteen months late. What happened?

The ZBA hearing in late 2027 drew the abutters who had raised open-land and traffic concerns since 2024; approval was conditioned on a redesigned screening plan and a smaller footprint, which sent the civil package back through Haugland in the winter of 2028. Borings, finally drilled in spring 2027, found debris deeper than the 2014 closure set showed, and the platform went from 2.5 ft of fill to a piled solution; the HEG reconciliation that had never been done was done as a change order. Nomad's twelve containers arrived on schedule in January 2028 and sat in a rented yard for nine months with the support clock running. Station 496 cutovers needed night outage windows priced at straight time. AFUDC alone added \$4M. The full-funding PAF said Conceptual, -25/+50%. Everyone remembered \$57M.

Early-warning indicators

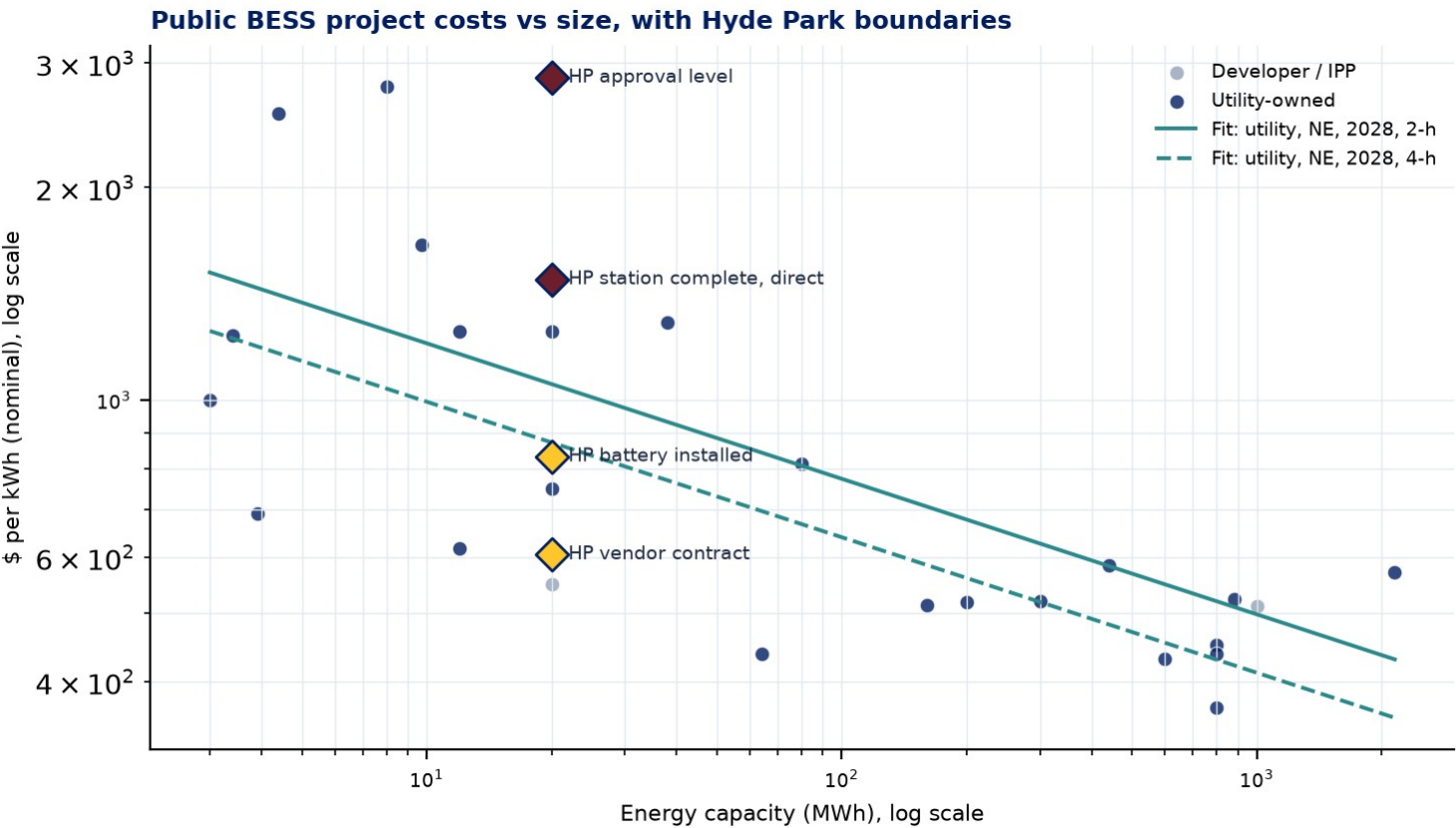
Indicator	Caught today?
Geotech report not issued by IFC-civil design freeze (mid-2027)	No: no P6 gate ties IFC to it
ZBA hearing date not set within 90 days of Article 80 filing	No: siting carries hours, not milestones
HEG reconciliation not closed before the PAF is drafted (Nov 2026)	Only if the CoE makes it a precondition
Nomad final-drawings milestone (13 Jul 2026) slipped	Partly: payments tracked, not downstream effect on GSU spec and civil
Station 496 outage plan not issued before construction pricing	No

Thirteen utility-owned projects at or below 100 MWh: where Hyde Park sits



- **Median** \$1,235/kWh; interquartile \$750 to \$1,289.
- **Hyde Park battery installed, \$833:** 38% percentile.
- **Station complete, direct, \$1,483:** 77% percentile, beside Fairhaven (\$1,659 actual, 2023), Chateaugay (\$1,490, 2023) and Provincetown (\$1,289, 2022).
- **Approval level, \$2,862:** above every member. The loaders (\$23.7M) and the two interconnection scopes are what the comparables do not carry.
- **Caveat:** figures retrieved from search excerpts of regulator and utility documents; verify at source before external use.

Size and duration explain two thirds of the spread in public costs; Hyde Park's direct cost is high-normal

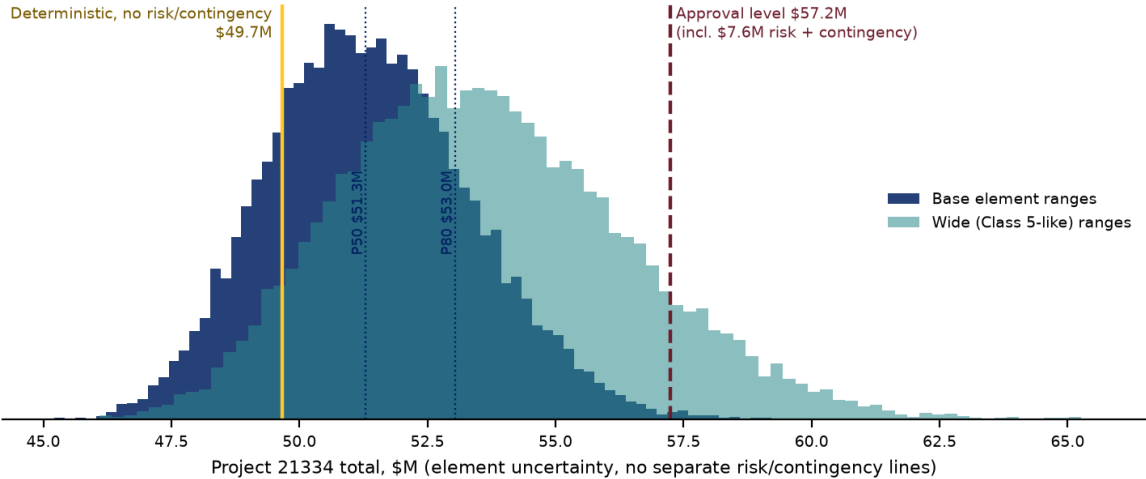


Term	Coefficient	p
Size (ln MWh)	-0.17	0.012
Duration (ln h)	-0.29	0.016
Year	+0.015	0.68
R-squared	0.65	n = 26

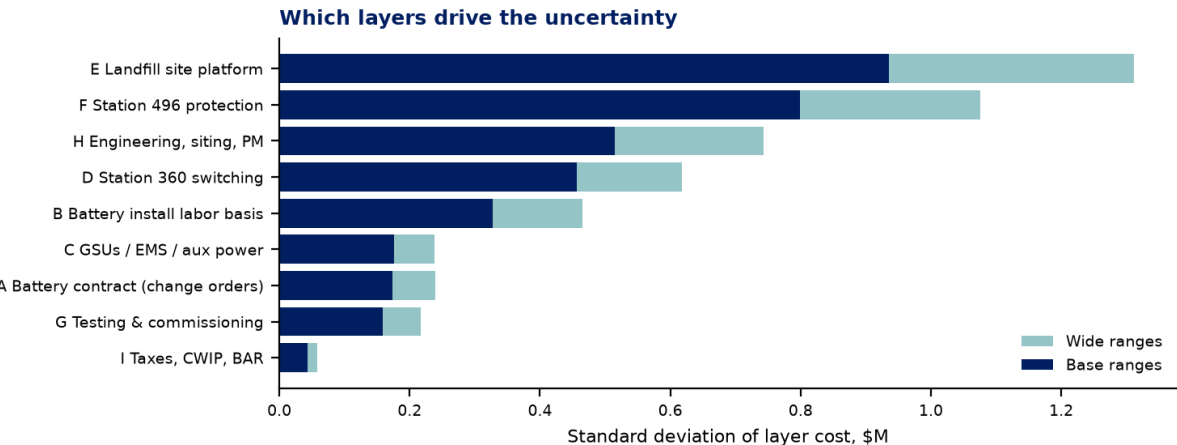
- **20 MWh, 2-h, 2028 prediction:** \$1,116/kWh, 80% interval \$611 to \$2,037.
- **Hyde Park percentiles:** contract 10%, installed 27%, station direct 73%, approval 98%.
- **Use as a bracket, not a price.** Residual spread is a factor of 1.4; ownership and region terms are not significant at this n.

Monte Carlo on the layers: the reserve covers the element ranges but is thin against outside history; the risk is in the ground and the station

Monte Carlo on the thirteen layers: where the risk and contingency reserve lands



	Base ranges	Wide ranges
P50 total	\$51.3M	\$53.4M
P80 total	\$53.0M	\$55.9M
Reserve to reach P80	\$3.4M	\$6.2M
Carried risk + contingency	\$7.6M	\$7.6M
P(total > \$57.2M)	0%	10%



- Distributions** are PERT ranges on each of the nine Station 360 layers plus Station 496, indirects and AFUDC; risk register and contingency are not stacked on top (they fund the same outcomes).
- Estimate-to-actual uplift on comparables:** Fairhaven +64%, Provincetown +9 to +40%, Waena +37%, Regina +31%, Hyde Park Sep cycle +21%. Median about +31%; Rev 3 carries 17.9%.
- Next step:** replace these ranges with the project team's in a two-hour range-estimating session (AACE 41R-08) before full funding.

CONTRIBUTIONS

A \$12 million battery is a \$57 million project, and now every dollar between them has a name

- **Cost layer extraction.** 292 Rev 3 line items in thirteen layers, reconciled to the penny. Direct shares: battery contract 23%, battery install and balance of plant 6%, station and interconnection 24%, owner scope 11%, Station 360 loaders 35%.
- **Unit costs at every boundary.** \$605/kWh vendor, \$833 installed, \$2,500 Station 360, \$2,862 project, \$3,102 all-in. Benchmarks compared against the row that matches their definition.
- **Cost history.** 2021 IFR to Rev 3, with the ISD drift and the five-week September walk documented from the ESF revisions and email record.
- **Red team and open-item register.** Three kill shots, seven stress tests, a pre-mortem with five early-warning indicators, and ten reconciliations with owners and dollars.

Decisions requested

- Adopt the layered view as the standard way this project is presented internally and to regulators: battery, station, interconnection, owner scope, loaders.
- Make the Haugland-to-Rev 3 reconciliation a precondition of the full-funding PAF; it is the one item that can move the base by more than the reserve.
- Present November at its real maturity (Class 4, -25/+50%) unless geotech, HMA and the HEG map close first.
- Put a float target on the siting chain and assign owners to the five early-warning indicators.
- Decide the escalation treatment on the fixed-price battery balance; retire the \$43.9M placeholder.